Mastering Decimals: Exercise 3C Solutions

Multiplication of Decimals — Complete Step-by-Step Solutions Guide

Working Rules for Multiplication of Decimals

Multiplication of decimals involves simple base-ten principles. The process depends on whether we are multiplying by a power of 10, a whole number, or another decimal. Below are the verified working rules from the curriculum textbook:

Rule 1: Multiplication by Powers of 10

To multiply a decimal by 10, 100, 1000, etc., shift the decimal point to the **right** by as many places as there are zeros in the multiplier. If necessary, annex zeros to the right of the decimal part to complete the shift.

Rule 2: Multiplication by Decimals/Whole Numbers

Multiply the numbers as whole numbers (ignoring the decimal points). In the resulting product, place the decimal point such that the number of decimal places is equal to the **sum of the decimal places** of the multipliers.

Question 1: Multiply (Multiplication by Powers of 10)

Apply **Rule 1**: Shift the decimal point to the right by 1 place for multiplying by 10, 2 places for 100, and 3 places for 1000.

Part	Expression	Shift Direction & Zeros	Product	Final Answer
(i)	6.5 × 10	Shift right 1 place (1 zero)	65	**65**
(ii)	0.8 × 10	Shift right 1 place (1 zero)	08	**8**
(iii)	9.07 × 10	Shift right 1 place (1 zero)	90.7	**90.7**
(iv)	2.345 × 10	Shift right 1 place (1 zero)	23.45	**23.45**
(v)	4.63 × 100	Shift right 2 places (2 zeros)	463	**463**
(vi)	8.279 × 100	Shift right 2 places (2 zeros)	827.9	**827.9**
(vii)	7.8 × 100	Shift right 2 places (annex one zero)	780	**780**
(viii)	0.09 × 100	Shift right 2 places (2 zeros)	009	**9**
(ix)	0.283 × 1000	Shift right 3 places (3 zeros)	283	**283**
(x)	6.25 × 1000	Shift right 3 places (annex one zero)	6250	**6250**
(xi)	5.4 × 1000	Shift right 3 places (annex two zeros)	5400	**5400**
(xii)	0.3 × 1000	Shift right 3 places (annex two zeros)	300	**300**

Question 2: Multiply (Decimal by Whole Number)

Apply **Rule 2**: Multiply as whole numbers, then count the decimal places in the decimal factor. The product must have the exact same number of decimal places.

Part	Expression	Whole Number Multiplication	Decimal Places	Final Product
(i)	2.4 × 16	24 × 16 = 384	1 place (in 2.4)	**38.4**
(ii)	3.45 × 17	345 × 17 = 5865	2 places (in 3.45)	**58.65**
(iii)	0.86 × 14	86 × 14 = 1204	2 places (in 0.86)	**12.04**
(iv)	2.68 × 30	268 × 30 = 8040	2 places (in 2.68)	**80.40 = 80.4**
(v)	0.023 × 65	23 × 65 = 1495	3 places (in 0.023)	**1.495**
(vi)	0.0006 × 15	6 × 15 = 90	4 places (in 0.0006)	**0.0090 = 0.009**

Question 3: Find the Product (Decimal by Decimal)

Apply **Rule 2** (Decimal by Decimal): Multiply both factors as whole numbers. The total decimal places in the product is equal to the sum of decimal places of both factors.

Part	Expression	Whole Multiplication	Decimal Places Sum	Final Product
(i)	5.6 × 1.4	56 × 14 = 784	1 + 1 = 2 places	**7.84**
(ii)	2.35 × 7.2	235 × 72 = 16920	2 + 1 = 3 places	**16.920 = 16.92**
(iii)	0.37 × 0.26	37 × 26 = 962	2 + 2 = 4 places	**0.0962**
(iv)	0.74 × 6.7	74 × 67 = 4958	2 + 1 = 3 places	**4.958**
(v)	5.64 × 0.08	564 × 8 = 4512	2 + 2 = 4 places	**0.4512**
(vi)	2.75 × 1.7	275 × 17 = 4675	2 + 1 = 3 places	**4.675**
(vii)	0.04 × 0.36	4 × 36 = 144	2 + 2 = 4 places	**0.0144**
(viii)	34.2 × 1.86	342 × 186 = 63612	1 + 2 = 3 places	**63.612**
(ix)	0.028 × 0.9	28 × 9 = 252	3 + 1 = 4 places	**0.0252**

Question 4: Find the Product (Three Decimals)

Multiply all three factors as whole numbers and place the decimal point such that the total decimal places equals the sum of the decimal places of all three numbers.

Part	Expression	Whole Number Product	Decimal Places Sum	Final Product
(i)	2.3 × 0.23 × 0.1	23 × 23 × 1 = 529	1 + 2 + 1 = 4 places	**0.0529**
(ii)	1.2 × 3.5 × 0.3	12 × 35 × 3 = 1260	1 + 1 + 1 = 3 places	**1.260 = 1.26**
(iii)	0.6 × 1.5 × 0.7	6 × 15 × 7 = 630	1 + 1 + 1 = 3 places	**0.630 = 0.63**
(iv)	0.2 × 0.2 × 0.02	2 × 2 × 2 = 8	1 + 1 + 2 = 4 places	**0.0008**
(v)	1.1 × 0.1 × 0.11	11 × 1 × 11 = 121	1 + 1 + 2 = 4 places	**0.0121**
(vi)	0.6 × 0.06 × 0.006	6 × 6 × 6 = 216	1 + 2 + 3 = 6 places	**0.000216**

Question 5: Evaluate (Squares and Cubes of Decimals)

Squaring a number means multiplying it by itself. Cubing a number means multiplying it three times. We count total decimal places accordingly (2 times the decimal places for square, 3 times for cube).

Part	Expression	Mathematical Expansion	Total Decimal Places	Final Value
(i)	$(1.3)^2$	1.3 × 1.3 = 1.69	1 × 2 = 2 places	**1.69**
(ii)	$(0.06)^2$	0.06 × 0.06 = 0.0036	2 × 2 = 4 places	**0.0036**
(iii)	$(0.2)^3$	0.2 × 0.2 × 0.2 = 0.008	1 × 3 = 3 places	**0.008**
(iv)	$(0.8)^3$	0.8 × 0.8 × 0.8 = 0.512	1 × 3 = 3 places	**0.512**

Word Problems: Step-by-Step Practical Applications

Question 6: The cost of one pen is Rs. 42.25. Find the cost of one dozen such pens.

• **Given:** Cost of 1 pen = Rs. 42.25

• **Requirement:** Find the cost of 1 dozen pens. (Recall: 1 dozen = 12 items)

• **Calculation:** Total Cost = Cost per pen × Quantity = 42.25 × 12

• **Arithmetic Step:** Multiply 4225 × 12 = 50700. Since 42.25 has 2 decimal places, place the decimal point 2 positions from the right: 507.00

Final Answer: Rs. 507 (or Rs. 507.00)

Question 7: A car moves at a constant speed of 56.4 km per hour. How much distance does it cover in 3.5 hours?

• **Given:** Speed of the car = 56.4 km/h, Time duration = 3.5 hours

• **Formula:** Distance Covered = Speed × Time = 56.4 × 3.5

• **Arithmetic Step 1:** Multiply as whole numbers: 564 × 35 = 19740

• **Arithmetic Step 2:** Total decimal places: 1 (in 56.4) + 1 (in 3.5) = 2 places. Place the decimal point 2 positions from the right: 197.40

Final Answer: 197.4 km (or 197.40 km)

Question 8: A room is 4.5 m long and 3.8 m broad. Calculate the area of the floor of the room.

• **Given:** Length of the floor (l) = 4.5 m, Breadth of the floor (b) = 3.8 m

• **Formula:** Area of the rectangular floor = Length × Breadth = 4.5 × 3.8

• **Arithmetic Step 1:** Multiply as whole numbers: 45 × 38 = 1710

• **Arithmetic Step 2:** Total decimal places: 1 (in 4.5) + 1 (in 3.8) = 2 places. Place the decimal point 2 positions from the right: 17.10

Final Answer: 17.1 sq m (or 17.10 m^2)

Question 9: The cost of 1 litre of refined oil is Rs. 124.75. What is the cost of 6.2 litres of this oil?

• **Given:** Cost of 1 litre of refined oil = Rs. 124.75, Quantity of oil = 6.2 litres

• **Calculation:** Total Cost = Cost per litre × Quantity = 124.75 × 6.2

• **Arithmetic Step 1:** Multiply as whole numbers: 12475 × 62 = 773450

• **Arithmetic Step 2:** Total decimal places: 2 (in 124.75) + 1 (in 6.2) = 3 places. Place the decimal point 3 positions from the right: 773.450

Final Answer: Rs. 773.45